

AMENDMENT NO. 1 JANUARY 1995
TO
IS 13349 : 1992 CAST IRON SINGLE FACED
THIMBLE MOUNTED SLUICE
GATES — SPECIFICATION

(*Second cover page, Foreword, second para, fourth line*) — Substitute '1 200' for '12'.

[*Second cover page, Foreword, fourth para, (b), first line*] — Delete the word 'gate' between 'back' and 'frame'.

(*Second cover page, Foreword, fifth para, fourth line*) — Substitute 'used' for 'use'.

(*Second cover page, Foreword, eighth para, fourth line*) — Substitute the words 'relating to' after the words 'the practices'.

(*Page 2, Table 3*) — Substitute the following for the existing:

Table 3 Materials
(*Clause 6.1*)

Sl No.	Item	Material	Conforming to IS	Grade/Designation
(1)	(2)	(3)	(4)	(5)
1.	Thimble, frame, guide, extension guide, slide (shutter), gear box, pedestal, stem guide, bracket, wedging devices, flush bottom seal support bgr	Cast Iron	210 : 1993	FG 200 or higher
2	Yoke (Bridge)	Cast Iron Structural Steel	210 : 1993 2062 1992	FG 200 or higher Grade A
3	a) Wedges	Cast Iron Naval Brass Phosphor Bronze Leaded Tin Bronze Stainless Steel	210 : 1993 291 : 1989 28 1985 318 : 1981 3444 : 1987	FG 200 or higher Grade 1 or Grade 2 Grade 1 or Grade 2 LTB 1 or LTB 2 Grade 2
	b) Wedge facings	Naval Brass Phosphor Bronze	291 : 1989 7814 1985	Grade 1 or Grade 2 HB, Grade 2 or HB, Grade 3

Price Group 1

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Table 3 Materials (Continued)

Sl No.	Item	Material	Conforming to IS	Grade/Designation
(1)	(2)	(3)	(4)	(5)
		Leaded Tin Bronze	318 : 1981	LTB 1 or LTB 2
		Stainless Steel	6911 : 1992	04Cr1Ni or 04Cr17Ni12Mo2Ti or 12Cr12 or 15Cr16Ni2
4	Operating nut and Guide Bush	Leaded Tin Bronze	318 : 1981	LTB 1 or LTB 2
5	Connecting block	Cast Iron	210 : 1993	FG 200 or higher
		Tin Bronze	306 : 1983	Sand Cast
		Leaded Tin Bronze	318 : 1981	LTB 1 or LTB 2
6	Seating faces	Naval Brass	291 : 1989	Grade 1 or Grade 2
		Phosphor Bronze	7814 : 1985	HB, Grade 2 or HB, Grade 3
		Tin Bronze	306 : 1983	Sand Cast
		Leaded Tin Bronze	318 : 1981	LTB 1 or LTB 1
		Stainless Steel	6911 : 1992	04Cr1, Nine or 04Cr17Ni12Mo2Ti or 12Cr12 or 15Cr16Ni2
7	Stem extension rod	Mild Steel	2062 : 1992	Grade A
		Stainless Steel	6603 : 1972	04Cr18Ni10 or 04Cr17Ni12Mo2Ti20 or 12Cr13 or 15Cr16Ni2
8	Flush bottom resilient seal	Natural or Synthetic Rubber	11855 : 1986	
9	Coupling	Cast Iron	210 : 1993	FG 200 or higher
		Mild Steel	2062 : 1992	Grade A
		Stainless Steel	6603 : 1972	04Cr18Ni10 or 04Cr17Ni12Mo2Ti20 or 12Cr13 or 15Cr16Ni2
10	Flush bottom seal retainer bar	Cast Iron	210 : 1993	FG 200 or higher
		Phosphor Bronze	7814 : 1985	HB, Grade 2 or HB, Grade 3
		Stainless Steel	6603 : 1972	04Cr18Ni10 or 04Cr17Ni12Mo2Ti20 or 12Cr13 or 15Cr16Ni2

Table 3 Materials (Concluded)

Sl No.	Item	Material	Conforming to IS	Grade/Designation
(1)	(2)	(3)	(4)	(5)
11	Anchor bolts	Mild Steel Stainless Steel	2062 . 1992 6603 . 1972	Grade A 04Cr18Ni10 or 04Cr17Ni12Mo2Ti20 or 12Cr13 or 15Cr16Ni2
12	Fasteners like assembly bolts, studs, nuts, etc.			
	a) For wall thimbles	Stainless Steel	6603 1972	04Cr18Ni10 or 04Cr17Ni12Mo2Ti20
	b) For gate assembly, stem guides and lifting devices	Mild Steel	1363 (Part 1) 1992 and 1363 (Part 3) 1993 6603 1972	04Cr 18Ni10 or 04Cr17Ni12Mo2Ti20

NOTES

1 In case of brass, bronze or stainless steel used for various components, only the chemical composition as specified in the relevant Indian Standard shall be complied with

2 When copper zinc alloy comes in contact with water containing acid or alkali, zinc from the alloy dissolves leaving the alloy metal porous and weak. This loss of zinc from the copper zinc alloy results in type of corrosion called dezincification. Low zinc content may resist dezincification

3 Use of stainless steel for the seating faces and wedges should be avoided wherever possible, because stainless steel machined surfaces rubbing against each other under high pressure have a tendency to gall which may result in severe damage to the sliding surfaces.

The use of brass should be avoided where dezincification can occur. Bronze having less than 2 percent aluminium and 5 percent zinc should be used where dezincification of bronze can occur. If dezincification of the bronze is a problem, slide gate seat facings and bronze castings should be phosphor or silicone bronze both of which meeting the low zinc requirement

(Page 4, clause 7.5.2.4) — Delete.

(Page 4, clause 7.8.1, ninth line) — Delete the word 'to'.

(Page 5, clause 7.11.3, first line) — Substitute the word 'furnished' for 'furnishes'

(Page 5, clause 7.12.1, fourth line) — Substitute the word 'hold' for 'held'.

(Page 5, clause 7.12.2, eleventh line) — Substitute the word 'with' for 'be'.

(Page 5, clause 8.1, eighth line) — Substitute the word ‘use’ for ‘used’.	30
(Page 5, clause 9.1, seventh line) — Substitute the word ‘mechanism’ for ‘machanism’.	31
(Page 5, clause 9.2, fifth line) — Substitute the word ‘bearing’ for ‘hearing’ and ‘enclosed’ for ‘erclosed’.	206
(Page 6, clause 9.3, tenth line) — Substitute the words ‘on which’ for ‘or which’.	344
(Page 6, clause 9.3.2, second line) — Substitute the word ‘manner’ for ‘marner’.	660
(Page 6, clause 9.4, third line) — Substitute the word ‘crank’ for ‘crark’.	691
(Page 7, clause 12.1, second para, sixth line) — Add the word ‘of’ between the words ‘warping’ and ‘the gate’.	781
(Page 7, clause 13.3.2, second line) — Add the word ‘for’ between ‘leakage’ and ‘class 1’.	1122
(Page 8, clause A-1, third line) -- Substitute ‘data’ for ‘date’.	1185
(Page 8, clause A-1, Sl No. 12) — Add the word ‘Field’ before ‘Leakage’.	
(Page 8, clause A-1, Sl No. 14) — Substitute the following for the existing:	‘life’
‘14. Wedges : Whether the wedging devices should have cast iron, bronze or stainless steel wedges. If wedges are of cast iron, whether they should have cast iron contacting faces or should have brass or stainless steel facings.’	
(Page 9, Annex B) — Substitute the following for the existing:	

‘ANNEX B

(Clause 2.1)

LIST OF REFERRED INDIAN STANDARDS

<i>IS No.</i>	<i>Title</i>	
28 : 1985	Specification for phosphor bronze ingots and castings (<i>second revision</i>)	
210 : 1993	Specification for grey iron castings (<i>fourth revision</i>)	(C
291 : 1989	Specification for naval brass rods and sections (suitable for machining and forging) (<i>second revision</i>)	

e' for	306	1983	Specification for tin bronze ingots and castings (<i>third revision</i>)
necha-	318	: 1981	Specification for leaded tin bronze ingots and castings (<i>second revision</i>)
earing'	2062	1992	Specification for steel for general structural purposes (<i>fouth revision</i>)
which'	3444	· 1987	Specification for corrosion resistant high alloy steel and nickle base castings for general applications (<i>second revision</i>)
anner'	6603	: 1972	Specification for stainless steel bars and flats
nk' for	6911	1992	Specification for stainless steel plate, sheet and strip (<i>first revision</i>)
rd 'of'	7814	: 1985	Specification for phosphor bronze sheet and strip (<i>first revision</i>)
etween	11228	· 1985	Recommendations for design of screw hoists
e'.	11855	: 1986	General requirements for rubber seals for hydraulic gates'
before	(Page 9, clause C-3, second line) — Substitute the word 'lift' for		
for the	'life'.		
st iron,			
st iron,			
· should			
sting'			

castings

ision)
(suitable

(CED 3)